Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *CIVIL COMMENTS* | ***Context:*** *Gazipur Peri-Urban Low-Literacy Health Chatbot Users (Bangladesh)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output modality — text-in, continuous-score-out via threshold-agnostic AUC and AEG metrics [Q11, Q14] — is broadly compatible with a content-moderation pipeline that scores incoming chatbot messages. The threshold-agnostic design [Q7, Q8, Q15, Q17] is methodologically sound and explicitly motivated by the observation that threshold-dependent metrics 'can obscure the view of unintended bias' [Q8]. Multi-label continuous scores in the actual data [DATASET-D1, D2, D3] confirm structural compatibility. However, score calibration was derived from English-language online comment distributions, and the paper itself cautions 'the correct handling of subtle variations in distributions must be decided on case by case basis' [Q35] and 'it's always possible that there is subtle bias that the metrics cannot detect' [Q36]. For Banglish/code-switched input, the underlying classifier scores would themselves be unreliable, meaning the metrics would measure noise rather than signal — but the metric framework itself is reusable if applied to a re-calibrated, re-annotated Bangladeshi evaluation set.

ID	Evidence
Q7	“We propose a suite of threshold agnostic performance metrics to measure the extent of unintended model bias. Many prior methods for measuring unintended bias in classification systems rely on selecting a threshold, a choice that can drastically change results.” (p.1)
Q8	“For these models, threshold dependant metrics can obscure the view of unintended bias and thus be misleading to practitioners. Threshold agnostic metrics capture the behavior of the underlying model itself, and thus can allow a more comprehensive comparison of the model's performance and limitations.” (p.1)
Q11	“We therefore propose a suite of five metrics, derived from ROC-AUC, Equality Gap, and Mann-Whitney U metrics, each of which captures a different aspect of model performance, and a different potential type of unintended bias.” (p.2)
Q14	“Most metrics for unintended bias rely on dividing the test data up by identity or demographic based subgroups and computing metrics for each group. For our metrics, we also divide data by subgroup. However, instead of calculating metrics on the subgroup data exclusively, our metrics compare the subgroup to the rest of the data, which we call the "background" data.” (p.2)
Q15	“A core benefit of AUC is that it is threshold agnostic.” (p.2)
Q17	“An important quality of the AUC metric is that it is robust to data imbalances in the amount of negative and positive examples in the test set.” (p.3)
Q35	“It's important to note that the correct handling of subtle variations in distributions must be decided on case by case basis.” (p.5)
Q36	“And, as with any other suite of metrics, it's always possible that there is subtle bias that the metrics cannot detect.” (p.5)
D1	You're an idiot. — Clear direct insult, English, standard register
D2	Yet call out all Muslims for the acts of a few will get you pilloried. So why is it okay to smear an entire religion over these few idiots? Or is this because it's okay to bash Christian sects? — Religious identity mention with high toxicity/identity_attack scores — Western Christian/Muslim framing
D3	haha you guys are a bunch of losers. — Casual insult, no identity dimension

Strengths

- Threshold-agnostic AUC-based metric suite (Subgroup AUC, BPSN AUC, BNSP AUC, Positive AEG, Negative AEG) [Q11] is structurally reusable for a Gazipur audit and explicitly designed to be robust to class imbalance [Q17, Q19] — a property highly relevant given expected toxicity-rate imbalance across Bangladeshi identity subgroups.
- Multi-label continuous scoring architecture (toxicity, identity_attack, insult, obscene, threat) provides finer-grained output structure than a binary label, supporting deeper bias diagnostics [DATASET-D1, D2, D3].

ID	Evidence
Q11	“We therefore propose a suite of five metrics, derived from ROC-AUC, Equality Gap, and Mann-Whitney U metrics, each of which captures a different aspect of model performance, and a different potential type of unintended bias.” (p.2)
Q17	“An important quality of the AUC metric is that it is robust to data imbalances in the amount of negative and positive examples in the test set.” (p.3)
Q19	“Enforcing that for each AUC, either all negative or all positive examples (or both in Subgroup AUC) come from one identity group, means that mis-orderings involving that particular subset cannot be drowned out by results from other groups, ensuring that these metrics are robust to data imbalances likely to occur in real data.” (p.3)
D1	You're an idiot. — Clear direct insult, English, standard register
D2	Yet call out all Muslims for the acts of a few will get you pilloried. So why is it okay to smear an entire religion over these few idiots? Or is this because it's okay to bash Christian sects? — Religious identity mention with high toxicity/identity_attack scores — Western Christian/Muslim framing
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality matches: deployment requires text-in, score-out toxicity classifier output for content-moderation gating; benchmark evaluates the same modality [Q11, Q70].

OF-2: Text-to-speech is not a benchmark feature; deployment is text-only per region YAML interaction_modality, so this is not a relevant gap.

OF-3: Literacy is highly relevant: BBS functional literacy is 62.92% nationally (7+) [WEB-2], with peri-urban industrial workers likely lower. The benchmark's score-out form supports backend gating decisions but does not address whether classifier rejection messages are accessible — this is an integration concern beyond the benchmark's scope.

OF-4: External validity is partially preserved at the form level (text-in/score-out matches), but score calibration mismatch [Q35] means absolute scores and threshold choices derived from this benchmark will not transfer to deployment input distribution. Documented external validity caveat by the authors themselves [Q35, Q36].

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Q11	“We therefore propose a suite of five metrics, derived from ROC-AUC, Equality Gap, and Mann-Whitney U metrics, each of which captures a different aspect of model performance, and a different potential type of unintended bias.” (p.2)
Q35	“It's important to note that the correct handling of subtle variations in distributions must be decided on case by case basis.” (p.5)
Q36	“And, as with any other suite of metrics, it's always possible that there is subtle bias that the metrics cannot detect.” (p.5)
Q70	“We introduced a new suite of metrics for unintended bias, based on ROC-AUC and Mann-Whitney U scores.” (p.8)
WEB-2	BBS functional literacy 62.92% (7+) — relevant context for downstream accessibility but not a direct OF benchmark gap (https://www.dhakatribune.com/bangladesh/education/319301/bbs-functional-literacy-rate-7-above-years-in)
WEB-4	Urban functional literacy 80.35%, rural 70.54% — peri-urban Gazipur likely between these (https://pressxpress.org/2023/07/18/bangladesh-bureau-of-statistics-reveals-long-strides-forward-in-literacy-rate/)

Information Gaps

- Score-calibration analysis on Banglish inputs has not been performed by anyone published

Requires Expert Verification

- Engineering verification of pipeline integration and threshold-choice strategy under deployment input distribution

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Remediation

Gap 1: Score calibration was derived from English distributions and is not validated for Banglish input [Q35]

Recommendation: Re-instantiate the threshold-agnostic AUC/AEG metric suite on the new Bangladeshi-annotated evaluation set; preserve the structurally reusable methodology while replacing all calibration anchors. Report metrics stratified by input-form degradation level and by Bangladeshi-relevant identity subgroups.

ID	Evidence
Q35	"It's important to note that the correct handling of subtle variations in distributions must be decided on case by case basis." (p.5)